

1. Assertion – Chloroplasts usually align themselves along the walls of mesophyll cells.

Reason – They get optimum quantity of incident light by aligning along well.

Choose the correct option.

(1) Assertion and Reason are correct and Reason is correct explanation for Assertion

(2) Assertion and Reason are correct but is not the explanation of Assertion

(3) Assertion and Reason are both incorrect

(4) Assertion is correct but Reason is incorrect

2. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Gymnosperm seeds are naked.

Reason (R) : They lack ovary wall.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

3. Match column I with column II and choose the right option

Column I	Column II
A. Ascomycetes	(i) Rhizopus
B. Basidiomycetes	(ii) Penicillium
C. Deuteromycetes	(iii) Ustilago
D. Phycomycetes	(iv) Alternaria

- (1) A-(iv), B-(iii), C-(i), D-(ii)**
- (2) A-(ii), B-(iii), C-(iv), D-(i)**
- (3) A-(iv), B-(i), C-(ii), D-(iii)**
- (4) A-(iii), B-(iv), C-(ii), D-(i)**

4. Given below are two statements :

Statement I : Meiotic division occurs generally in reproductive cells.

Statement II : Synapsis occurs during zygotene cells.

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect**
- (2) Statement I is correct but Statement II is incorrect**
- (3) Statement I is incorrect but Statement II is correct**
- (4) Both, Statement I and Statement II are correct**

5. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Bryophytes are amphibians of plant kingdom.

Reason (R) : In a bryophyte plant dependent on water for sexual reproduction.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)**
- (2) (A) is correct but (R) is not correct**
- (3) (A) is not correct but (R) is correct**
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)**

6. Match the following

a. Polyadelphous i. Brinjal

b. Epipetalous ii. Citrus

c. Epiphyllous iii. Lily

d. Perigynous iv. Peach

(1) a-ii, b-i, c-iii, d-iv

(2) a-i, b-ii, c-iii, d-iv

(3) a-iv, b-ii, c-iii, d-i

(4) a-ii, b-iv, c-iii, d-i

7. Match the following :

- | | |
|---------------------------------|-----------------------|
| (A) Human urine | (i) Ethylene |
| (B) Coconut milk | (ii) GA3 |
| (C) Ripened oranges | (iii) Auxin |
| (D) Gibberella fujikuroi | (iv) Cytokinin |

(1) A - iii, B - iv, C - i, D - ii

(2) A - i, B - ii, C - iii, D - iv

(3) A - iv, B - iii, C - ii, D - i

(4) A - ii, B - i, C - iv, D - iii

8. How many NADH + H⁺ will be produced during the production of molecule of acetyl CoA from two molecule of pyruvic acid :

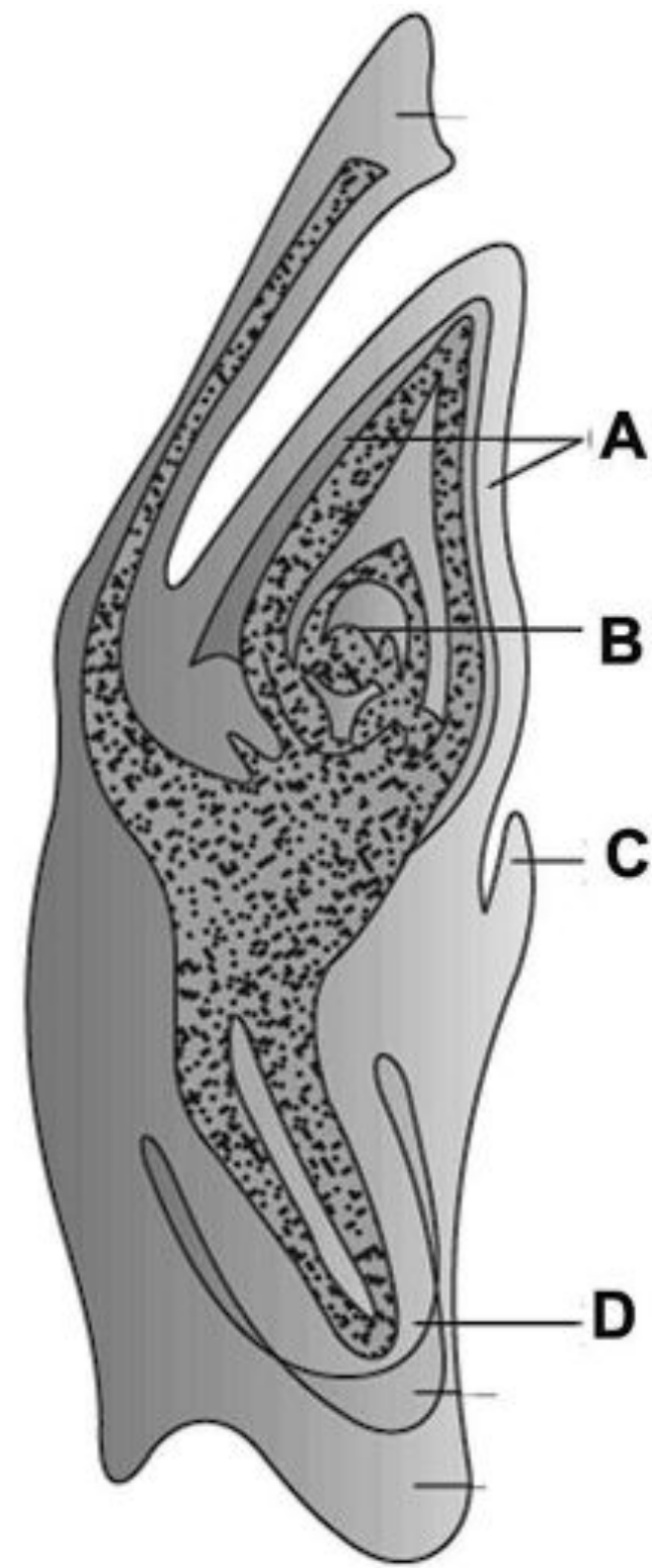
(1) 1 NADH + H⁺

(2) 2 NADH + H⁺

(3) 3 NADH + H⁺

(4) 4 NADH + H⁺

9. Label the A, B, C and D :



- (1) A-Shoot apex, B-Coleoptile, C-Epiblast and D-Radicle**
- (2) A-Coleoptile, B-Epiblast, C-Shoot apex and D-Radicle**
- (3) A-Coleoptile, B-Shoot apex, C- Radicle and D-Epiblast**
- (4) A-Coleoptile, B-Shoot apex, C-Epiblast and DRadicle**

10. Consider the following statements.

(a) Phloem fibres are made up of sclerenchymatous cells

(b) Phloem parenchyma are generally present in monocots

(c) The first formed primary phloem consists of narrow sieve tube

and is referred to as protophloem and the later formed phloem

has bigger sieve tubes and is referred to as metaphloem

Which statement(s) is/are false ?

(1) Only a

(2) Only b

(3) Only c

(4) a, b, and c

11. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : C₄ plant also show CO₂ fixation by Calvin - Benson cycle.

Reason (R) : Mesophyll cells in C₄ plants show Calvin cycle CO₂ fixation.

In the light of the above statements, choose the correct answer from the options given below

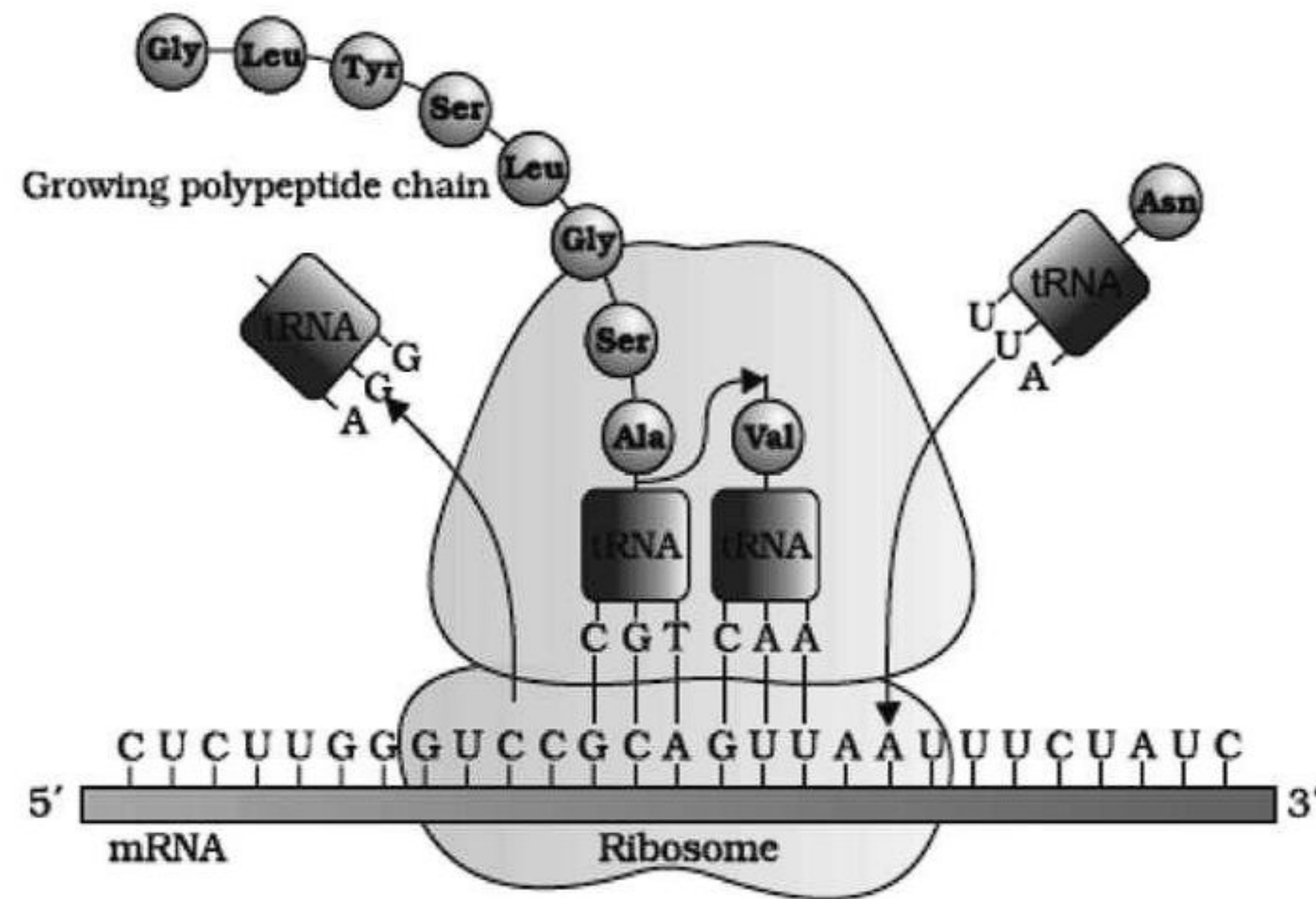
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

12. With reference to the given below diagram below depicting a process taking place in a bacteria, the enzyme that would catalyze a peptide bond between Ala and Val is



- (1) 23S rRNA**
- (2) 28S rRNA**
- (3) Peptidyl phosphorylase**
- (4) Peptidyl dehydrogenase**

13. Match the following columns :

Column – I

Column – II

(a) RNA polymerase –I

(i) tRNAs

(b) RNA polymerase–II

(ii) rRNAs

(c) RNA polymerase–III

(iii) 5srRNAs

(iv) Sn RNAs

(v) hn RNAs

(1) a-ii, b-i, iii, iv, c-v

(2) a-ii, b-v, c-i, iii, iv

(3) a-v, b-i, iii, iv, c-ii

(4) a-i, b-ii, c-iii, iv

14. Which one of the following experiment in which radioactive elements/compound are used :

- (a) Frederick griffith experiments**
- (b) Hershey and Chase experiments**
- (c) Meselson and stahl experiments**
- (d) Taylor experiments**
- (e) T.H. morgan experiments**

(1) a, b, c, d, e

(2) b, d

(3) b, c, d

(4) Only b, c

15.Sclereids are present in

(A) Fruit wall of nuts

(B) Pulp of guava, pear and sapota

(C) Seed coat of legume

(1) Only A

(2) Only B

(3) Only C

(4) All A, B and C

16. Which one of the following is not a Mendelian disorder:

(1) Cystic fibrosis

(2) Thalassemia

(3) A disorder with characteristic palm crease

(4) Phenylketonuria

17. Sickle cell anaemia is caused due to substitution of:

(1) Glutamic acid by valine at the sixth position of chain

(2) Valine by glutamic acid at the sixth position of chain

(3) Glutamic acid by valine at the sixth position of chain

(4) Valine by glutamic acid at the sixth position of chain

18. Which of the following characters of Drosophila is not suitable for genetical studies ?

(a) They could be grown on simple synthetic medium in laboratory

(b) They complete their life cycle in about 2-weeks

(c) Single mating produces few number of progeny flies.

(d) They have many types of hereditary variations that can be seen with low power microscope.

(e) Male & Female flies are not easily distinguishable

(1) a, b, c

(2) a, b, c, d, e

(3) d and e

(4) c and e

19. $dN/dt = rN$ equation is related to

(a) Exponential growth

(b) Geometric growth

(c) J. Shaped curve

(d) $N_t = N_o e^{rt}$

(1) only a, b

(2) only b, c

(3) only a, d

(4) All of these

20. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): An ecosystem can be visualised as a functional unit of nature.

Reason (R) : Living organisms interact among themselves and also with the surrounding physical environment in ecosystem.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

21. Match the following columns :

Column – I	Column – II
(a) Mango	(i) On site conservation
(b) National Park	(ii) Ex-situ conservation
(c) Wildlife Safari Park	(iii) 34
(d) Hotspots in world	(iv) 1000 varieties
(1) a-iv, b-i, c-iii, d-ii	
(2) a-iv, b-i, c-ii, d-iii	
(3) a-iv, b-ii, c-i, d-iii	
(4) a-ii, b-i, c-iii, d-iv	

22. Select the incorrect pair among the followings :

(1) Dodo – New Zealand

(2) Thylacine – Australia

(3) Quagga – Africa

(4) Steller's sea cow – Russia

23. There is a need to standardise the naming of living organisms is known by the same name all over the world. This process is called :

(1) Classifications

(2) Nomenclature

(3) Both 1 and 2

(4) identifications

24. Select the correct statements for mendelian disorder

(a) It may be dominant or recessive

(b) It may be autosomal or sex-linked

(c) In mendelian disorder the number of chromosomes are not changed

(d) The mendelian disorders are mainly determined by alteration or mutation in the single gene

(1) All

(2) Only a, b, c

(3) Only b, c, d

(4) Only a, b, d

25. Which statement is correct :

(a) Biologists are not sure about how many prokaryotic species there might be

(b) India is among one of the 12 mega diversity countries of the world

(c) Species diversity increases as we move away from equator towards poles.

(d) Amazonia rain forest is home to about 3000 species of fishes.

(1) a, b only

(2) a, b, c only

(3) a, d only

(4) a, b, d only

26. The Gnathostomata are divided into two super classes, this classification is based on which character :

(1) Presence of type of appendages

(2) Presence of type of heart

(3) Presence of type of kidney

(4) Presence of type of bone

27. Match the following:

A. Hippocampus i. Fighting fish

B. Beta ii. Sea horse

C. Clarias iii. Rohu

D. Labeo iv. Magur

(1) A–ii; B–i; C–iv; D–iii

(2) A–iii; B–i; C–iv; D–ii

(3) A–i; B–ii; C–iv; D–iii

(4) A–i; B–iv; C–ii; D–iii

28. Match Column- I with Column- II and select the correct option :

Column - I

Column-II

a. Cuboidal

i. Wall of blood vessels

b. Ciliated

ii. Lining of stomach and intestine

c. Columnar

iii. Inner lining of fallopian tube

d. Squamous

iv. Duct of glands

(1) a -i, b-ii, c-iii, d- iv

(2) a -ii, b-iii, c-i, d- iv

(3) a -iv, b-iii, c-ii, d- i

(4) a -iii, b-iv, c-i, d- ii

29. The body of cockroach divisible into how many distinct regions :

(1) 3

(2) 4

(3) 2

(4) 20

30. During extreme summer frog take shelter in deep burrows to protect them, this one called :

(1) Hibernation

(2) Aestivation

(3) Mimicry

(4) Protection

31. Correct statement about lipid :

- a. These are generally water insoluble**
- b. Lecithin is a Phospholipid**
- c. Glycerol is trihydric alcohol**
- d. Phospholipid are found in cell membrane**

(1) Only a, b, c, d

(2) Only a, c and d

(3) Only c, b, d

(4) Only c, d

32. Given below are two statements :

Assertion (A) : Cellulose can not hold I_2 .

Reason (R) : Cellulose does not contain complex helices.

Choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct

explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of

(A)

33. Which of the following are aromatic amino acids

(a) Tryptophan

(b) Tyrosine

(c) Phenylalanine

(d) Histidine

(1) all of these

(2) a,b,c

(3) a and b only

(4) b and c only

34. Match column - I with column - II and select the correct option from the codes given below :

Column - I

A. TV + ERV

B. RV + ERV + TV + IRV

C. ERV + RV

Column - II

(i) Expiratory capacity

(ii) Total lung capacity

(iii) Functional residual capacity

(1) A - (i), B - (ii), C - (iii)

(2) A - (iii), B - (i), C - (ii)

(3) A - (iii), B - (ii), C - (i)

(4) A - (ii), B - (iii), C - (i)

35. Given below are two statements :

Assertion (A) : Thrombins in turn are formed from inactive substance present in plasma called prothrombin.

Reason (R) : An enzyme complex fibrinogen required for changing prothrombin to thrombin.

Choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

36. Assertion (A): Cerebellum has convoluted surface

Reason (R): Cerebellum has many more neurons

(1) Assertion and Reason both are correct and Reason is correct explanation of Assertion.

(2) Assertion and Reason both are correct but Reason is not correct explanation of Assertion.

(3) Assertion is correct but Reason is wrong

(4) Assertion is wrong but Reason is correct

37. Clotting factors are present in

(a) plasma

(b) in inactive form

(c) in active form

(d) serum

(1) a and b

(2) a and c

(3) d and b

(4) d and c

38. Given below are two statements :

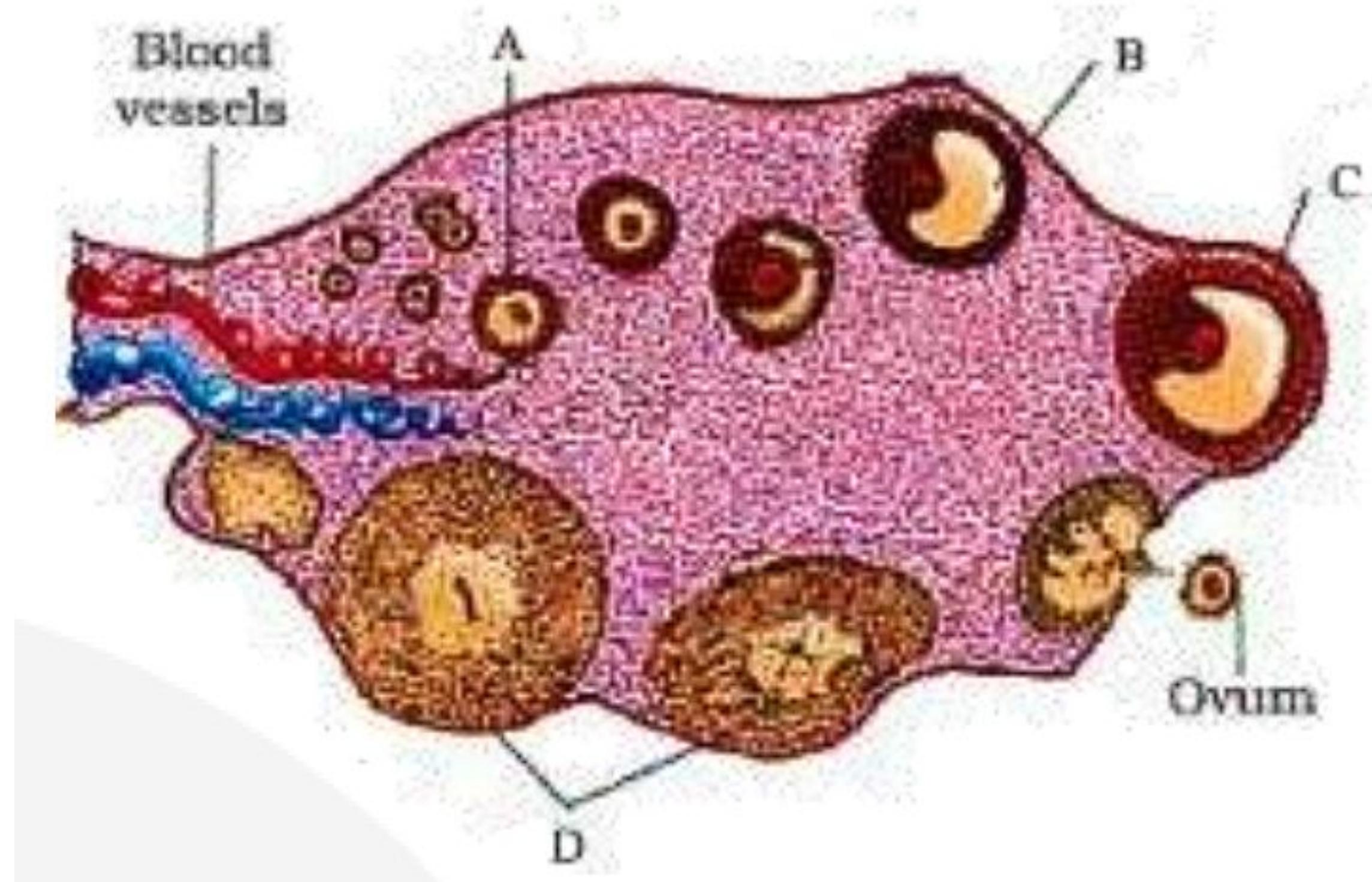
Statement I : The labia minora are paired folds of tissue under labia majora.

Statement II : Each spermatogonium is diploid and contains 23 chromosomes.

In the light of the above statements, choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect**
- (2) Statement I is correct but Statement II is incorrect**
- (3) Statement I is incorrect but Statement II is correct**
- (4) Both Statement I and Statement II are correct**

39. The figure given below shows the sectional view of ovary. Select the option which gives correct identification of marked structure (A to D) and its feature



- (1) A: Primary follicle, it is also called gamete mother cell.**
- (2) B: Corpus luteum, it cannot be formed and added after birth.**
- (3) C: Graafian follicle, mature follicle which ruptures to release secondary oocyte.**
- (4) D: Tertiary follicle, a large number of this follicle degenerates during the phase from birth to puberty**

40. Given below are two statements :

Statement I : The world population which was around 2 billion in 1900.

Statement II : The world population were 7.2 billion in 2000.

In the light of the above statements, choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect**
- (2) Statement I is correct but Statement II is incorrect**
- (3) Statement I is incorrect but Statement II is correct**
- (4) Both Statement I and Statement II are correct**

41. Given below are two statements :

Statement I : Reproductive events occur before puberty.

Statement II : Sperm formation continues even in old man.

**In the light of the above statements, choose the correct answer
from the options given below**

- (1) Both Statement I and Statement II are incorrect**
- (2) Statement I is correct but Statement II is incorrect**
- (3) Statement I is incorrect but Statement II is correct**
- (4) Both Statement I and Statement II are correct**

42. Given below are two statements :

Statement I : The universe is very old almost 20 billion years old.

Statement II : In the solar system of the milky way galaxy, earth was supposed to have been formed about 4.5 billion years back.

In the light of the above statements, choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect**
- (2) Statement I is correct but Statement II is incorrect**
- (3) Statement I is incorrect but Statement II is correct**
- (4) Both Statement I and Statement II are correct**

43. Given below are two statements :

Statement I : The green revolution succeeded in tripling the food supply.

Statement II : 'GMO' stands for genetically modified organisms.

In the light of the above statements, choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect**
- (2) Statement I is correct but Statement II is incorrect**
- (3) Statement I is incorrect but Statement II is correct**
- (4) Both Statement I and Statement II are correct**

44. Match Column - I with Column - II

Column - I	Column - II
(A) Spiders	(i) Gills
(B) King crab	(ii) Book lungs
(C) Prawn	(iii) Books gills
(D) Labeo	

- (1) A-(iii), B-(i), C-(ii), D-(ii)
- (2) A-(ii), B-(iii), C-(i), D-(i)
- (3) A-(i), B-(ii), C-(iii), D-(ii)
- (4) A-(i), B-(iii), C-(i), D-(ii)

45. Match the following :

- | | |
|-----------------|---|
| (a) IUT | (i) Embryo transfer in fallopian tube at 1-8 celled |
| (b) ZIFT | (ii) Embryo transfer in uterus at more than 8 celled |
| (c) GIFT | (iii) Induced abortion |
| (d) MTP | (iv) Ova transferred in fallopian tube. |

(1) a-ii, b-i, c-iv, d-iii

(2) a-i, b-ii, c-iii, d-iv

(3) a-iv, b-i, c-iii, d-iv

(4) a-i, b-iv, c-iii, d-ii

46.Statement-I : Trachea is a straight tube extending upto the mid-Abdominal cavity, which divides at the level of 5th Lumbar vertebra into a right and left primary bronchi.

Statement-II : The trachea, primary, secondary and tertiary bronchi and initial bronchioles are supported by incomplete cartilaginous rings.

- (1) Both Statement-I and II both are incorrect.**
- (2) Statement-I is correct but statement-II is incorrect.**
- (3) Statement-I is incorrect but Statement-II is correct.**
- (4) Both Statement-I and Statement-II are correct.**

47. Choose the correct option for wuchereria?

(I) Triploblastic and presence of an excretory pore

(II) Presence of a muscular pharynx

(III) Males longer than females

(IV) Cellular level of organization

(1) (II) and (III) are true

(2) (I) and (IV) are true

(3) (I) and (II) are true

(4) (III) and (IV) are true

48. Evolution in nature could possible due to:

a – Natural selection

b – Gene recombination

c – Mutation

d – Genetic drift

e – Gene flow

(1) a and c only

(2) b, c, d and e

(3) a, c, d and e

(4) a, b, c, d and e

49. Descending loop of henle is :

a. Permeable to water

b. Impermeable to electrolytes

c. Impermeable to water

d. Permeable to electrolytes

(1) a, b are true

(2) c,d are true

(3) a,d are true

(4) b,c are true

50. Given below are two statements :

Statement I : Chondrichthyes are marine water but migrate for spawning to fresh water.

Statement II : Chondrichthyes have 6–15 pairs of gill slits for respiration which are covered by an operculum

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect**
- (2) Statement I is correct but Statement II is incorrect**
- (3) Statement I is incorrect but Statement II is correct**
- (4) Both Statement I and Statement II are correct.**

SOLUTION

1-1	11-2	21-2	31-1	41-3
2-4	12-1	22-1	32-4	42-4
3-2	13-2	23-2	33-2	43-4
4-4	14-2	24-1	34-1	44-2
5-1	15-4	25-4	35-2	45-1
6-1	16-3	26-1	36-1	46-3
7-1	17-1	27-1	37-1	47-3
8-2	18-4	28-3	38-2	48-4
9-4	19-4	29-1	39-3	49-1
10-2	20-4	30-2	40-2	50-1